Troubleshooting Guide: Cathodic Protection System — Rising Pipeline Corrosion Rate Despite Normal Rectifier Output

Category: Integrity Management / Corrosion Control **Unit:** Buried Pipeline — Impressed Current Cathodic Protection (ICCP) System **Tools:** Pipe-to-soil potential (P/S) survey, rectifier output trending, coating condition assessment (DCVG/CIS survey) **Fluid Package:** Not applicable — this is an electrochemical corrosion protection investigation, not a process simulation **Symptom:** Corrosion coupon/ER probe data showing an increasing external corrosion rate on a buried pipeline segment, despite the cathodic protection rectifier operating at its normal output

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1. Quick Reference

Item	Detail
Reported symptom	Increasing external corrosion rate detected on a buried pipeline segment, despite the CP rectifier delivering its normal, unchanged output current/voltage
Initially unclear	Whether the CP system itself was underperforming (undersized/failing) or something specific to this segment was preventing adequate protection from reaching the pipe surface, despite normal overall rectifier output
Actual root cause	A section of disbonded/damaged pipeline coating was “shielding” that segment of pipe from the impressed current, so even though overall rectifier output was normal and pipe-to-soil potential looked adequate at test stations, the actual steel surface under the disbonded coating was not receiving protective current
Fix	Excavated and repaired the coating in the affected section; verified direct-current-voltage-gradient (DCVG) or close-interval survey (CIS) confirmed restored protection at that specific location
Diagnostic signal	Overall rectifier output and nearby test station pipe-to-soil potentials looked normal, but a close-interval or DCVG survey specifically over the affected segment identified a localized potential shift consistent with coating shielding
Prevention	Periodic DCVG/CIS survey coverage (not just test station point readings) to catch localized shielding; coating condition assessment tied into the CP monitoring program

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2. Symptom

- **Corrosion coupon/electrical resistance (ER) probe data showed an increasing external corrosion rate** on a specific buried pipeline segment.
- The **cathodic protection rectifier was operating at its normal, unchanged output** current and voltage — the most obvious CP system health indicator looked fine.

3. Why This Wasn't Assumed to Be a Rectifier/CP System Capacity Issue

Rising corrosion rate on a cathodically protected pipeline is often first suspected to mean the CP system itself is underperforming — insufficient current output, a failing rectifier, or a system that's become undersized as the pipeline ages or soil conditions change. But here, rectifier output was already confirmed normal, and standard test station pipe-to-soil potential readings (typically taken at fixed intervals) also looked acceptable. This meant the problem, if real, had to be **localized** — something preventing protective current from reaching the steel at this specific segment, invisible to point measurements taken elsewhere along the line.

4. Diagnostic Approach

Step 1 — Confirm rectifier output and nearby test station readings

Rectifier current/voltage output was confirmed normal, and pipe-to-soil potential readings at the **nearest fixed test stations** were reviewed and found within the acceptable protection criteria — consistent with a functioning CP system overall.

Step 2 — Recognize the limitation of point-based test station data

Fixed test stations only sample pipe-to-soil potential at specific points, often spaced considerable distances apart. A **localized** issue between test stations — such as a section of shielded coating — would not necessarily be visible in that data, even with a functioning CP system.

Step 3 — Run a close-interval or DCVG survey over the affected segment

A **close-interval survey (CIS)** or **direct-current-voltage-gradient (DCVG) survey** was conducted specifically over the pipeline segment showing the rising corrosion rate, providing continuous (rather than point-sampled) potential/current data along that section.

Finding: The survey identified a **localized potential shift** consistent with **coating shielding** — a section of disbonded or damaged coating that was preventing the impressed current from effectively reaching the underlying steel, even though the surrounding soil and pipe-to-soil potential looked adequate.

Step 4 — Confirm via excavation

The identified section was excavated, confirming **disbonded/damaged coating** at that specific location, consistent with the survey's shielding indication.

Quantitative Basis

- ER probe corrosion rate: baseline 1.2 mpy, rose to **6.8 mpy** against an acceptable threshold of <3 mpy for CP-protected pipe.
- Rectifier output held steady at **8.2 A / 22 V DC**, well within its 10 A/30 V rating — no change over the period.

- Nearest fixed test stations (spaced ~1,600 ft apart) both read **−0.95 V and −1.02 V CSE**, more negative than the −0.85 V NACE protection criterion — both indicating adequate protection by the standard point-based check.
- A close-interval survey at 2.5 ft spacing identified a localized shift to **−0.72 V CSE over an 18 ft section** roughly midway between the two test stations — less negative than the −0.85 V criterion, indicating inadequate protection at that specific location. Excavation confirmed **18 ft of disbonded/tented coating** matching the survey findings almost exactly.

5. Root Cause

A section of disbonded/damaged pipeline coating was shielding the underlying steel from the impressed cathodic protection current. Because overall rectifier output was normal and the nearest fixed test stations happened to be positioned outside the affected segment, this localized shielding was not visible through standard point-based CP monitoring — allowing corrosion to progress at this specific location despite an apparently healthy CP system overall.

6. Corrective Action

1. **Excavated and repaired the coating** in the affected section.
2. **Verified restored protection** at that specific location via a follow-up DCVG/CIS survey.

7. Verification

- Follow-up CIS survey over the repaired 18 ft section measured potential restored to **−0.91 V CSE**, better than the −0.85 V protection criterion.
- ER probe corrosion rate at the affected segment declined from 6.8 mpy to **1.4 mpy over the following 6 months** of monitoring — back within the acceptable <3 mpy range.

8. Prevention / Long-Term Fix

- Established **periodic DCVG/CIS survey coverage** along the pipeline route, rather than relying solely on fixed test station point readings, specifically to catch localized shielding issues that point-based monitoring can miss.
- Integrated **coating condition assessment** into the ongoing CP monitoring program, recognizing that coating and CP system performance are interdependent, not separately-managed parameters.

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9. General Troubleshooting Checklist (Reusable)

- ☐ Confirm rectifier output and nearby fixed test station pipe-to-soil potentials before assuming a CP system capacity/failure issue
- ☐ Recognize that normal rectifier output and normal nearby test station readings do NOT rule out a localized problem between test stations
- ☐ Run a close-interval survey (CIS) or DCVG survey specifically over the segment showing the elevated corrosion rate, for continuous rather than point-sampled data
- ☐ Look for localized potential shifts consistent with coating shielding or disbondment
- ☐ Confirm via excavation before committing to a repair scope
- ☐ Repair the identified coating section and verify restored protection with a follow-up survey at that specific location

- ☐ Confirm corrosion rate recovery via coupon/ER probe monitoring, not just the electrical survey result alone
- ☐ Build periodic CIS/DCVG survey coverage and coating condition assessment into the standing CP monitoring program, rather than relying only on fixed test station readings

10. Key Takeaway

> A cathodic protection system can be functioning perfectly at the rectifier and still fail to protect a specific pipeline segment, if disbonded coating is shielding that section from the impressed current. Fixed test station readings only sample specific points and can easily miss a localized shielding problem between stations — when corrosion rate rises despite normal rectifier output and normal nearby test station data, run a close-interval or DCVG survey specifically over the affected segment rather than concluding the CP system needs to be resized or upgraded.

Related Concepts / Tags

`cathodic-protection` `impressed-current` `coating-shielding` `disbonded-coating` `pipe-to-soil-potential` `DCVG` `close-interval-survey` `CIS` `pipeline-corrosion` `integrity-management`

This guide is derived from a representative field troubleshooting scenario. Values and thresholds shown are illustrative and should be validated against your own unit's design basis before applying.